

DAG Pipeline Declared by `pipeline.yaml` |

Single-project pipelines read `infrastructure/core/pipeline/pipeline.yaml`. `scripts/runner/execute_pipeline.py` expands the declarative DAG, applies tag filters (`--core-only` skips llm stages), checkpoints between nodes, then dispatches the declared `scripts/pipeline/stage_NN_*.py` scripts or builtin methods (`_run_clean_outputs`).

The **default YAML graph contains ten named core stages** (plus telemetry configuration metadata):

1. **Clean Output Directories** — wipes prior `projects/<name>/output/` + delivered `output/<name>/` paths so stale PDFs cannot satisfy validation.
2. **Environment Setup** (`scripts/pipeline/stage_00_setup.py`) — Python/uv probing, toolchain discovery, scaffolding directories, `PYTHONPATH` wiring.

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3. **Infrastructure Tests** (scripts/pipeline/stage_01_test.py --infra-only) — tests/ suite with the configured 60% source-coverage floor.
4. **Project Tests** (scripts/pipeline/stage_01_test.py --project-only) — isolated per-project suites with each project's declared floor (90% for this exemplar).
5. **Project Analysis** (scripts/pipeline/stage_02_analysis.py) — lexicographically ordered projects/<name>/scripts/*.py, each a thin orchestrator (src/ does real work).
6. **PDF Rendering** (scripts/pipeline/stage_03_render.py) — Pandoc → XeLaTeX loop, bibliography assembly, injected variables from Stage 02 artefacts.
7. **Output Validation** (scripts/pipeline/stage_04_validate.py) — PDF structure, manifests, Markdown hygiene.

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8. **LLM Scientific Review** (scripts/pipeline/stage_06_llm_review.py --reviews-only; tags: llm) — executive + quality critiques via local Ollama; allow_skip: true.
9. **LLM Translations** (scripts/pipeline/stage_06_llm_review.py --translations-only; tags llm, same dependency edges) — multilingual abstract expansion.
10. **Copy Outputs** (scripts/pipeline/stage_05_copy.py) — reproducible snapshots into canonical output/<project>/.

Two LLM nodes intentionally share one script module with orthogonal CLI switches; both depend only on validation so they can parallelize logically while remaining optional.

Executive reporting (scripts/pipeline/stage_07_executive_report.py) is **not** a YAML node inside the single-project executor. --all-projects / execute_multi_project.py invokes it once after iterating projects, consolidating cross-project KPIs dashboards.

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Topological order therefore differs slightly from declaration order (e.g., `copy` executes after `validation` even though `stage_05_copy.py` precedes `stage_06_llm_review.py` lexically).

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Stage Highlights

Infrastructure vs project tests. Splitting pytest invocations isolates flaky infra regressions (`MAX_TEST_FAILURES` knobs) from zero-tolerance gates on domain code (`max_project_test_failures` default 0 declared in YAML front-matter/testing blocks).

Stage 02 illustration. The analysis stage is deliberately concrete rather than a hypothetical diagram factory: each canonical project ships real behaviour at this node. `template_autoresearch_project` runs readiness validation; `template_search_project` merges remote literature JSON, generates scripted figures (`y_generate_search_figures.py`), and writes manifests; `template_code_project` emits optimization plots; and `template_prose_project` triggers structural validation scaffolding. The pipeline shape is identical across all four—only the Stage 02 payload differs—which is exactly what lets one orchestrator serve heterogeneous research domains.

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Interactive Orchestration

run.sh

Thin wrapper invoking `python -m infrastructure.orchestration`. Offers:

- ▶ per-project staged execution,
- ▶ chained digits (234 shorthand),
- ▶ multi-project grid (a-d presets),
- ▶ graceful quit / resume parity with `scripts/README.md`.

Selecting **d alone** after a passing multi-project run exits immediately once summaries print—avoiding repetitive menu redraw.

secure_run.sh

Executes Python secure path: standard pipeline artefact reproduction **then** invokes `run_secure_pipeline` for steganographic PDF hardening (`infrastructure.steganography`). Original PDFs stay immutable; hardened companions carry QR overlays plus hash manifests sidecars.